

Integrating AI Tools into Sponge City Concepts for Enhanced Flood Prevention and Mitigation: A Case Study of the Asahi River

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The core idea of sponge cities is to enhance urban water management by creating systems that absorb, store, and release water in a controlled way. These systems are designed to mitigate urban flooding as well as improve water quality and biodiversity. This paper explores how flood control elements within (sponge) cities could be optimized using advanced technologies like SAM, Hydro-STIV, and Random Forest. SAM was used for land cover classification, STIV and Hydro-STIV for estimating the flow velocity and Random Forest for water level prediction. The proposed methodology offers a path toward smarter and resilient cities.

Key Words: *Sponge City, Segment Anything Model, Hydro-STIV, Random Forest, Depth Anything V2*

1. INTRODUCTION

Urban flood management is becoming an increasingly urgent challenge as cities face rising pressures from climate change and rapid urbanization^{1), 2)}. The occurrence, duration, and frequency of flooding events are expected to increase under these circumstances. Traditional engineered flood defenses, such as levees, dikes, and stormwater drainage systems, are sometimes insufficient in coping with the intensity of extreme weather events. Consequently, cities are adopting new strategies like the “sponge city” concept, which relies on natural and semi-natural systems to absorb, store, and release stormwater gradually.

Synonyms for sponge city are green and blue infrastructure (GBI), low-impact development (LID), water sensitive urban design (WSUD), and sustainable flood risk management (SFRM).

The **sponge city** approach (**Fig.1**) provides benefits beyond flood control, including enhanced biodiversity, carbon reduction, and improved urban space quality³⁾. These cities incorporate features such as permeable surfaces, green roofs, rain gardens, urban parks and floodplains. Besides mitigating floods, these infrastructures can reduce carbon dioxide emissions, helping limit climate change and

thus heavy rains. Additionally, water stored in these systems can be released slowly to provide the resource during short dry periods.

While sponge city concepts have proven effective^{4), 5), 6), 7), 8)}, interest is growing in combining them with advanced technology to improve flood management. Artificial intelligence (AI) offers a powerful solution to enhance the efficiency of flood forecasting and urban water management. This paper focuses on three AI-based tools—Segment Anything Model (SAM), Hydro-Space Time Image Velocimetry (Hydro-STIV), and Random Forest (RF) model—evaluating their potential for flood mitigation (**Fig.2**).

Land Cover Classification (LCC) plays a crucial role in assessing the changing dynamics of land and its impact on flood behavior⁹⁾. Land cover influences infiltration and water storage capacities, as well as water levels due to surface roughness. Landcover mapping can therefore give insight into the “sponginess” of landscapes. Recently, AI tools like the **Segment Anything Model (SAM)**, developed by Meta AI, have shown potential in automating LCC through efficient image segmentation^{10), 11)}. SAM has the ability to segment objects using prompt-based and zero-shot capabilities, making it highly adaptable for broad geospatial analysis applications¹²⁾, such as

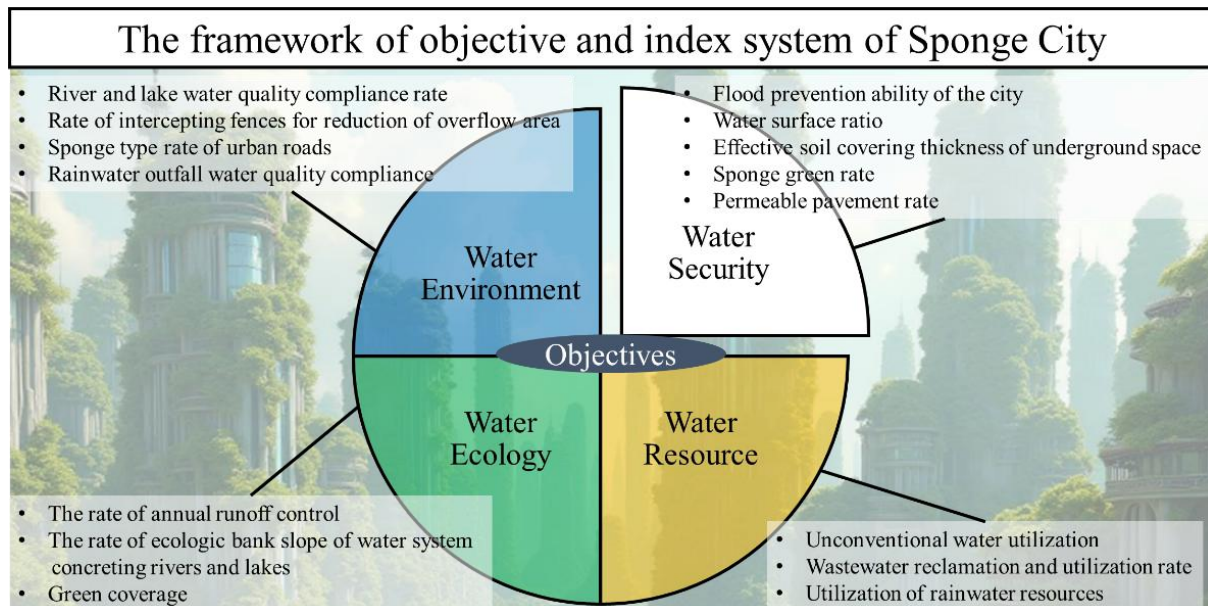


Fig.1 Sponge City Framework and the targeted sector in this paper ‘Water Security’ (adapted from ⁵⁾).

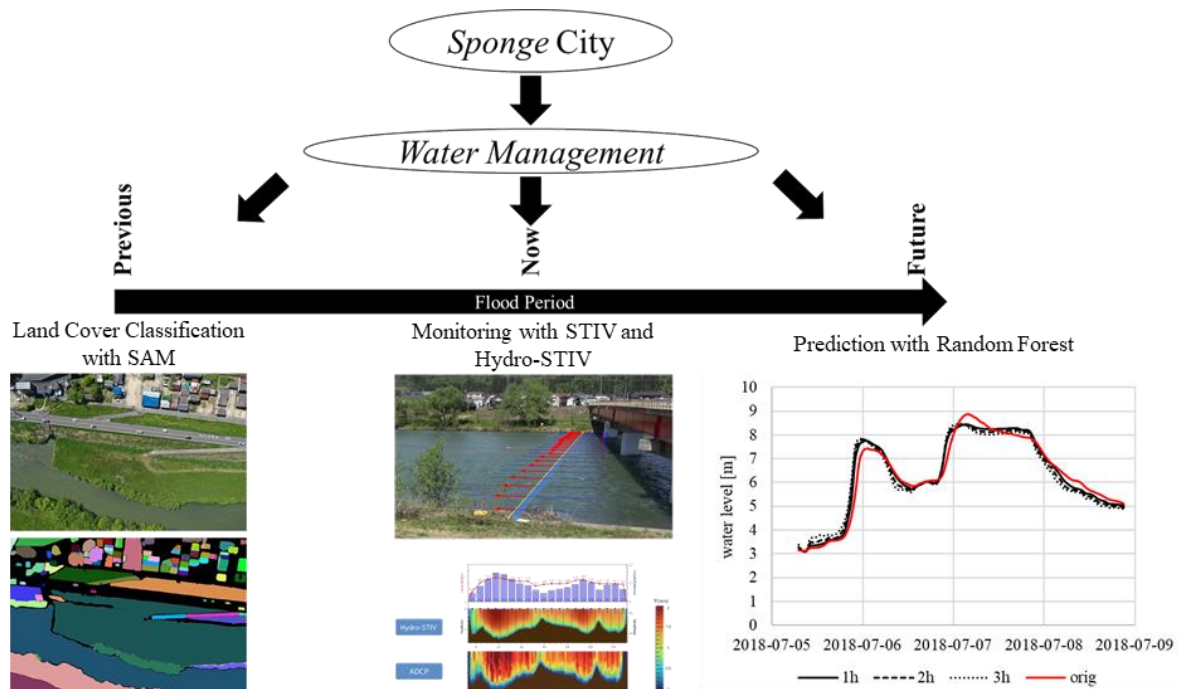


Fig.2 Graphical Abstract: Use aerial images from the past to test Land Cover Classification with SAM, discuss real-time discharge monitoring possibilities of Hydro-STIV, and do water level prediction with Random Forest.

LCC. Despite its advantages, SAM faces challenges in accuracy when tasked with fine details or subtle land cover (LC) differences.

Similarly, **Space-Time Image Velocimetry (STIV)** offers an advanced, non-contact approach for measuring river surface flows, using brightness variations or particles on the water surface to assess velocity¹³⁾. Image sources can be cameras of unmanned aerial vehicles (UAVs) or closed-circuit television (CCTV). The new AI-driven Hydro-STIV software enhances data processing speed and accuracy. This method offers an advantage over

traditional hydrological data collection techniques by providing continuous, high-resolution real-time information that can significantly enhance flood monitoring¹⁴⁾. STIV has been successfully applied in several studies^{15), 16), 17)}, e.g. showing similar accuracy as acoustic doppler current profiler (ADCP) measurements. Current research focuses on the main challenges like lighting and 2D flow. For instance, using far infrared (FIR) cameras has proven effective for measurement during night¹⁸⁾.

Random Forest (RF) is a highly acknowledged machine learning method in hydrology that is based

Table 1 Brief Overview of Sponge City Projects (SCP) examples in Japan, Germany and China. Classification of sponge city projects²⁾: (1) Contiguous open green spaces, interconnected waterways, channels and ponds, (2) Green roofs, (3) Porous design (e.g., permeable pavements and roads, bioswales), (4) Water savings and recycling (e.g., grey water).

Country	City (population in million)	Sponge City Projects
Japan	Tokyo (37)	(2), (3), (4) ¹³⁾ , 20), 21), 22)
	Kumamoto (0.7)	(3) ²²⁾
Germany	Berlin (3.6)	(1), (2), (3), (4) ²³⁾ , 24), 25), 26), 27)
	Hamburg (1.8)	(1), (2)
	Dresden (0.6)	(1), (3), (4) ²⁸⁾ , 29)
	Leipzig (0.6)	(1), (2), (3), (4) ³⁰⁾ , 31), 32), 33)
China (30 pilot cities)	Shanghai (30)	(1), (2), (3), (4) ³⁴⁾
	Guangzhou (15)	(1), (4) ³⁶⁾
	Shenzhen (13)	(1), (2), (3) ³⁴⁾ , 35)
	Wuhan (8.9)	(1), (3), (4) ³⁷⁾
	Jinan (5.9)	(1), (2), (3) ³⁸⁾

on decision trees. It has been widely used for flood and water level forecasting, achieving fast and accurate predictions³⁹⁾. It is a non-parametric method making it flexible for a wide range of data types. As data availability grows, RF's applications in flood hazard assessment are expected to expand. Schoppa *et al.* (2020)⁴⁰⁾ assume that it could become an alternative in large-scale flood hazard assessment.

This short literature review reveals that each method has already been used in the past for the same purposes described in this paper: SAM for LCC, STIV and Hydro-STIV for estimating river flow and RF for hydrological prediction. While each method was tested and evaluated individually, the novel points of this study are the combined use in the Asahi River basin with the idea of future application in a smart sponge city. Additionally, we introduced an improved LCC process with SAM including Depth Anything V2 (DA).

Some sponge city projects in Japan, Germany, and China are shown in **Table 1**. The Asahi River, for example, features a diversion weir that redirects excess discharge into a smaller riverbed with floodplains during high water levels. This technically makes it a sponge city element although Okayama doesn't designate it as such.

2. STUDY SITE AND METHODS

(1) Study site

This study focuses on the lower reaches of the Asahi River and its diversion into the Hyakken River in Okayama Prefecture, Japan. The selected site is the area around the Diversion weir, including a preceding flow diversion through the Myojo weir and a

vegetated island (**Fig.3**). The vegetated island is a key factor for the optimal discharge diversion regarding both rivers' capacities. The Asahi River is a Class I (state-controlled) river, with a catchment area of 1,810 km², and flows into the Seto Inland Sea⁴¹⁾. At Makiyama hydraulic station (20 KP), the mean discharge between 1965 and 2005 was 57.12 m³/s⁴²⁾. Historically, gravel bars lined this river reach, but the construction of dams upstream in the 1980s encouraged vegetation growth and reduced bedload transport¹⁴⁾.

The study site has experienced frequent flooding, with a notable event in July 2018 reaching a peak discharge of 4,500 m³/s costing lots of lives⁹⁾. The Asahi River and its diversion into the Hyakken River feature flood plains and floodable rice paddies. In sponge cities, extensive flood plains achieve the greatest possible effect on water level reduction. Additionally, the dual use of floodplains along the Hyakken River as different sports grounds is a good example of how floodplains can benefit the population. Apart from the flood control function, the study site is of particular interest due to the mix of urban settlements and different biotopes that support diverse wildlife.

(2) Land cover classification with SAM and SAM+DA

SAM was employed for LCC. We used the general segmentation configuration that works without any guiding prompts. As SAM segments all detectable objects without predefined classifications, it generated a random color for each object, requiring manual post-segmentation labeling.

Aerial images were collected from Google Earth Pro—provided by Maxar Technologies—across

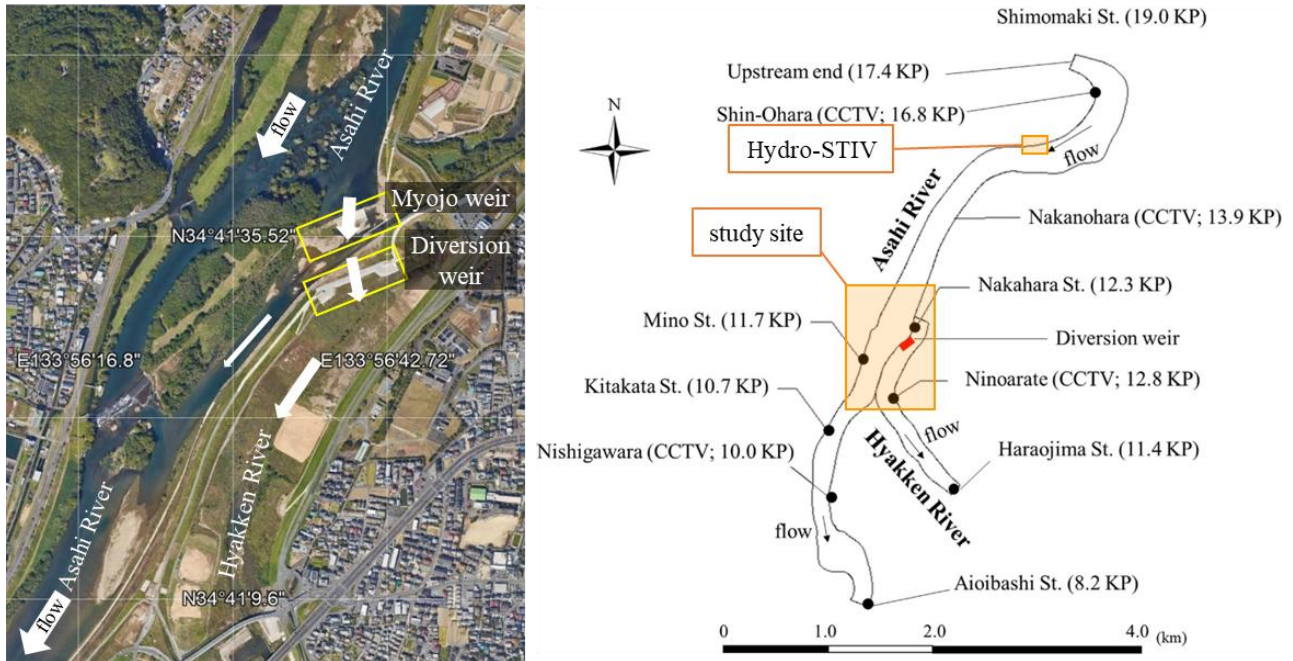


Fig.3 Study site around the Asahi River diversion weir (left) and measurement network with gauging stations and CCTVs (right).

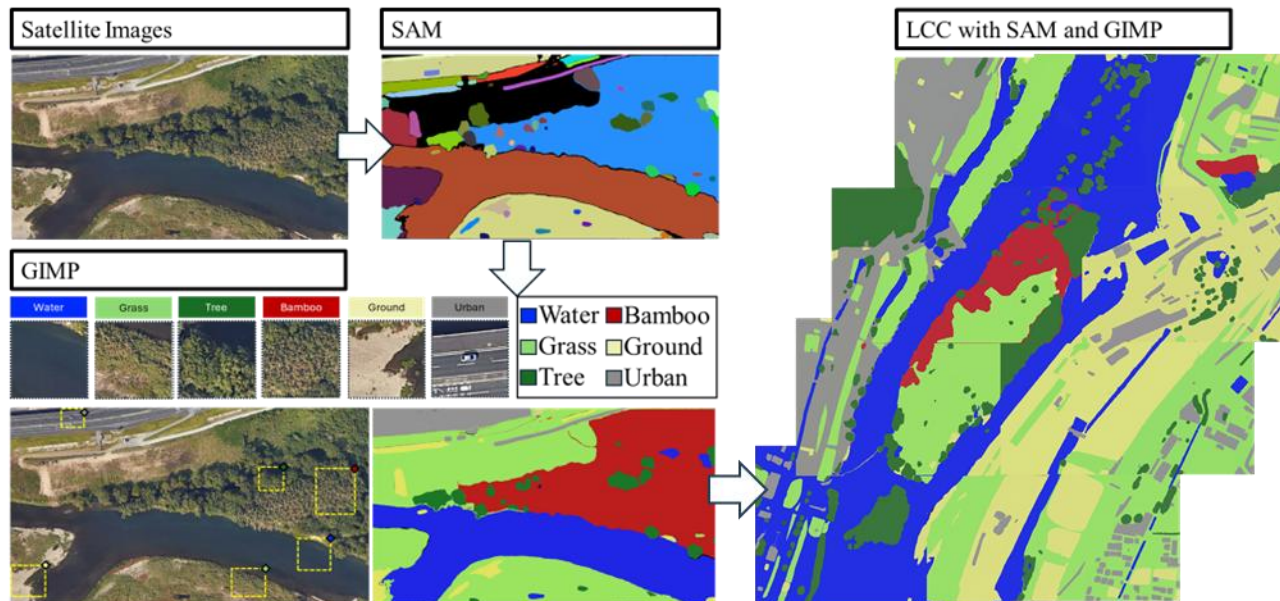


Fig.4 Flow chart of the Land Cover Classification (LCC) using the Segment Anything model (SAM) for automated segmentation and GIMP software for manual coloring.

various years and seasons, chosen for their broad accessibility and comparability (Sep 2009, Jan 2010, Feb 2016, May 2016, Apr 2018, Oct 2018, Mar 2021, Sep 2022, May 2024). Yet aerial imagery is limited by atmospheric and lighting conditions that could hinder segmentation accuracy. This study aimed to explore the potential for SAM to track riverine biotope changes over time and assess how seasonal vegetation differences and image quality impact segmentation.

GIMP software was used for the post-segmentation labeling (Water, Grass, Tree, Bamboo, Ground, Urban) by comparing the segmented images

with the original aerial images and if available UAV images from previous fieldwork (**Fig.4**). Urban means objects like houses, cars and roads.

A preliminary examination revealed poor segmentation when using large spatial aerial images covering almost the whole study site. Therefore, the area was divided into smaller aerial puzzle pieces (1912x1185, 96 dpi) to put together later (1655x1692, 330 dpi). In the case of ambiguous LC segments, a subjective decision was made either regarding the most present LC form or as a compromise. Disadvantage of the puzzle method are sharp edges when segmentation and thus LCC was not optimal.

In order to obtain better LCC results, we added a new component in the LCC process. the Depth Anything V2 demo for image depth analysis. In addition to the standard SAM-generated image, an additional SAM-generated image was produced after analyzing the aerial image for image depth. This led to two different segmentation results. Finally, both images were overlayed with transparency and merged in GIMP to identify more LC objects (**Fig. 5**).

True label mapping was done in GIMP by creating a new layer above each aerial image. Adjusting the transparency of the new layer allowed drawing LC polygons manually. The improvement was analyzed using confusion matrices that compared the color labels in the SAM-generated maps pixel by pixel with the true label maps. A high LC matching is indicated with a normalized percentage of 1 and bad matching with 0. If all LC types are matching well, a diagonal spans from the top left to the bottom right corner.

(3) Flow measurement with STIV and Hydro-STIV

Traditional contact-based methods for measuring river discharge face limitations in both accuracy and continuity, especially during flood events. Chen *et al.* (2024)⁴³ did an interesting review of different video-based non-contact river discharge measurement techniques. STIV, developed by Fujita *et al.* (2007)¹³, is a non-contact technique that captures natural ripples and corrugations on the water surface by camera instead of requiring tracer particles like particle tracking methods. The core assumption is that the brightness distribution on the river surface is correlated with the surface velocity. The typical STIV workflow involves four steps:

1. capturing water surface images with UAVs or CCTV,
2. extracting tracer information in a space-time image (STI),
3. calculating depth-averaged velocity by multiplying the surface velocity with the factor 0.85, and
4. summing subsection discharges to get the cross-section discharge.

To generate STIs, parallel and equal-length search lines are set up along the flow direction in each video frame to extract grayscale levels from these lines (**Fig.6**). The sequential arrangement of this data produces an STI where surface velocity is inferred by analyzing the angle between band textures in the image and the vertical axis, using methods like the Gradient Tensor Method (GTM) or Fast Fourier Transform (FFT). Recent developments like Hydro-STIV include a deep-learning-based convolutional neural network by Hydro Technology Institute Co., Ltd., making it almost fully automated⁴⁴. However, STIV is less suitable for 2D flows with varying directions, making it most effective in straight river sections.

This study incorporates both the traditional STIV and the new Hydro-STIV technique. Their application is contextualized by past and future flood events in the Asahi River. The 2018 flood, the most severe in recent years, is of special interest due to its intensity and the availability of STIV-compatible data. At the time of the event, Hydro-STIV was not yet developed, making traditional STIV the primary method for analyzing river flow with CCTV cameras. Future floods could be analyzed using Hydro-STIV. Until then, students practice using Hydro-STIV with drones, also under low flow conditions to familiarize themselves with its capabilities.

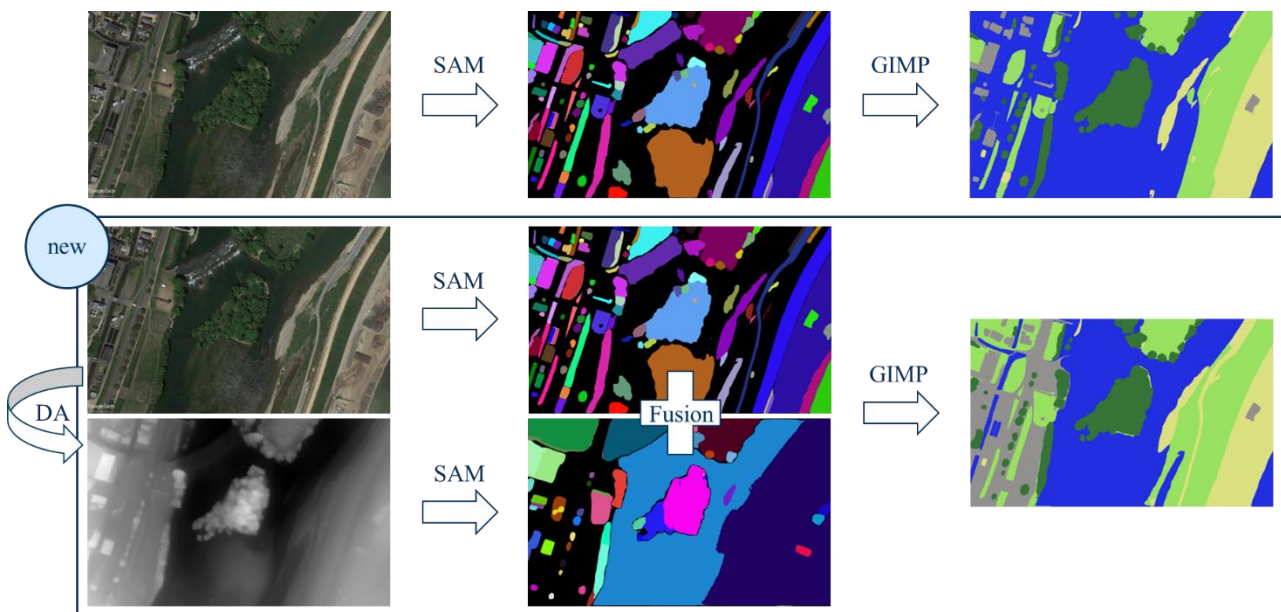


Fig.5 Flow chart of the improved LCC by including Depth Anything V2 (DA) in the segmentation process.

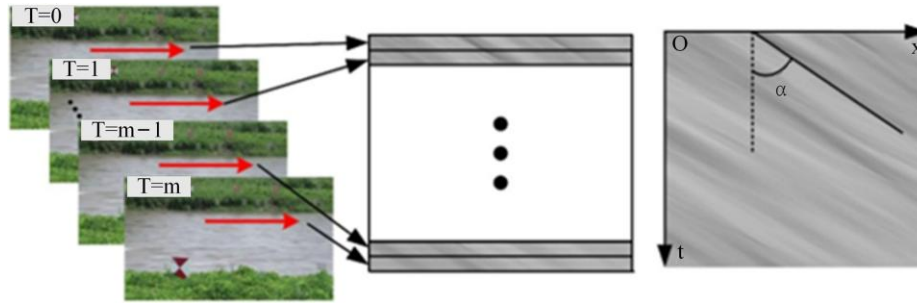


Fig.6 Generation of an STI for one search line (red arrow) and Identification of the texture gradient for calculating the surface velocity in the generated STI for the shown search line⁴³⁾.



Fig.7 Pictures taken during the drone Hydro-STIV fieldwork under low flow conditions in November 2024. The electromagnetic (EM) method served as an evaluation reference for the Hydro-STIV data.

STIV data was collected during the 2018 flood at Nakanohara, Nishigawara and Ninoarate among other operating CCTV HD cameras (**Fig.3** and **Fig.12**). The three sites are helpful to control if the discharge diversion ($Q_{\text{Nakanohara}} = Q_{\text{Nishigawara}} + Q_{\text{Ninoarate}}$) is optimal during flood events in order to initiate maintenance measures. For discharge calculation, water levels were estimated using water level bars on bridge piers or steps on the embankment opposite the cameras. At Nakanohara, 52 search lines with ~ 5.5 m intervals covered a river cross-section of nearly 300 m. At Nishigawara, 52 search lines with ~ 4.0 m intervals were set up, while Ninoarate had 36 search lines with ~ 6.3 m intervals for a river cross-section of over 200 m each⁴⁵⁾.

The peak discharge occurred shortly after 3 a.m. on July 7, 2018, under low light conditions. The Nakanohara CCTV camera changed the viewing angle after 24 a.m. when the discharge was still rising, showing only a part of the cross-section and thus limiting the usable video material. STIV analysis at Nishigawara and Ninoarate became feasible when it was getting bright. We chose to show the STIV from 6 a.m., where discharge was still high. For comparison, we used analytical flow velocity values from a 2D hydrodynamic model^{45), 46)}.

Hydro-STIV data was collected in a single river section in Gion area of the Asahi River (**Fig.3**) on October 30, 2024, using a DJI Mavic 2 Pro drone equipped with a camera (**Fig.7**). It flew in 50 m, 60 m, and 70 m height to get three different results for comparison. 17 search lines with 1m interval were

set on a clear to cloudy day with minimal wind. At the same site, the electromagnetic (EM) method was conducted as reference. Water level information was obtained during the EM.

(4) Water level forecast with RF

The Random Forest (RF) algorithm used in this study is a supervised learning model that constructs an ensemble of decision trees, each trained on a randomly selected subset of data. By averaging the predictions from each tree, RF minimizes overfitting and improves robustness, making it effective for predicting complex relationships within the dataset (**Fig.8**). This non-parametric model does not require assumptions about data distribution, allowing it to handle both categorical and continuous inputs like LCC and velocity data.

The study area is downstream Shimomaki and around Nakahara gauging stations on the Asahi River (**Fig.3**), where water level data from 13 flood events between July 2011 and July 2020 was collected. The objective was to predict water levels at Nakahara Station for each times step 1, 2, and 3 hours into the future using past water levels up to 5 hours (**Fig. 9**). The floods of September 2017, July 2018, and August 2021 were used as test cases. The model was trained utilizing data from all events except the target flood.

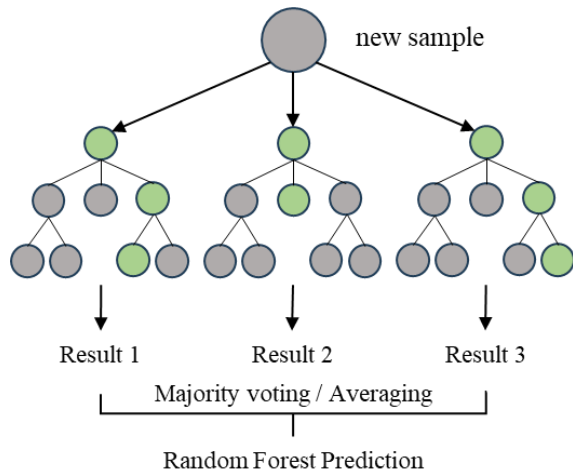


Fig.8 Understanding Random Forest (adapted from ⁴⁷).

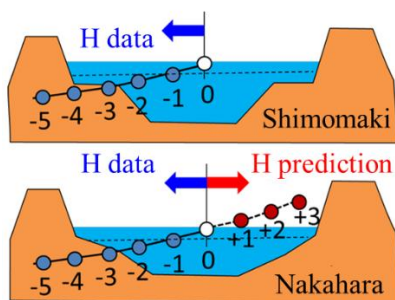


Fig.9 Water level (H) prediction (1-3 hours) for Nakahara Station using previous water level data (0-5 hours) from Shimomaki and Nakahara Station (Fig. 3).

3. RESULTS AND DISCUSSION

(1) Land cover classification with SAM and SAM+DA

The LCC results from SAM and GIMP were initially evaluated qualitatively by comparing the manually labeled segmented images with the original aerial images from Google Earth Pro (**Fig. 10**). As expected, higher-resolution aerial imagery produced a greater number of meaningful segments, highlighting the importance of fine-grained details in capturing the complexity of the Asahi River's landscape. Therefore, the mapping process involved using multiple aerial images from one timestamp as puzzle pieces to cover the entire study area.

In most cases, SAM successfully distinguished between water bodies, urban areas, ground, and vegetation, demonstrating its potential for environmental and geospatial analysis. However, a recurring issue throughout the process was the presence of large background segments that contained multiple LC types without sharp edges.

This indicates that SAM is optimized for identifying foreground objects or areas with high contrast, such as clusters of shadow-throwing trees,

houses, or bright ground, rather than large, relatively homogenous regions.

Furthermore, uncertainties in manual labeling emerged in cases where even the editing person struggled to differentiate land cover types due to similarity or image resolution and lighting. This highlights the inherent difficulty in using aerial imagery as ground truth. Manual labeling also proved to be time-intensive, suggesting that this approach for LCC with SAM may not be suitable for large-scale applications.

Later in the study, true label mapping (TL) was introduced to provide a valuable reference for a more profound evaluation of LCC. It allowed a quantitative comparison of the standard and the improved LCC method (SAM+DA), using confusion matrices for April and October 2018 (**Fig. 10** and **Fig. 11**).

In general, all confusion matrices display the expected diagonal pattern, indicating a reasonable level of accuracy. However, some misclassifications are notable, particularly in urban areas. In April 2018, SAM effectively identified water bodies (0.88) and bamboo (0.87), while urban areas were classified with lower accuracy (0.51).

A similar trend was observed in October 2018, where water bodies (0.89) were correctly labeled, but urban areas suffered from frequent misclassification as grass, leading to an even lower matching value (0.39). This intraannual discrepancy could be provoked by seasonal LC variations. The overall poor identification of urban areas was mainly caused by for SAM hardly distinguishable LC differences. Consequently, SAM created large background segments and thus it forced manual misclassification.

Another indication for seasonal LCC differences might be visible in the class Tree, as some deciduous trees had already shed their leaves by October, making them harder to distinguish for SAM, which in turn reduced tree classification matching compared to April.

The integration of DA significantly improved LCC accuracy in both April and October 2018, particularly benefiting urban area detection. In October 2018, urban classification accuracy increased from 0.39 to 0.68, while in April, it rose from 0.51 to 0.68. However, the ground classification accuracy in April slightly decreased, likely due to a manual labeling compromise issue: When poor segmentation occurred, the label editor had to make compromises, often selecting the most dominant land cover form. A vivid example of this is evident in the SAM and SAM+DA maps of October 2018, where a large water area appears on the right. Consequently, it can be assumed that the high matching percentage of water was partly achieved at the expense of other LC types.

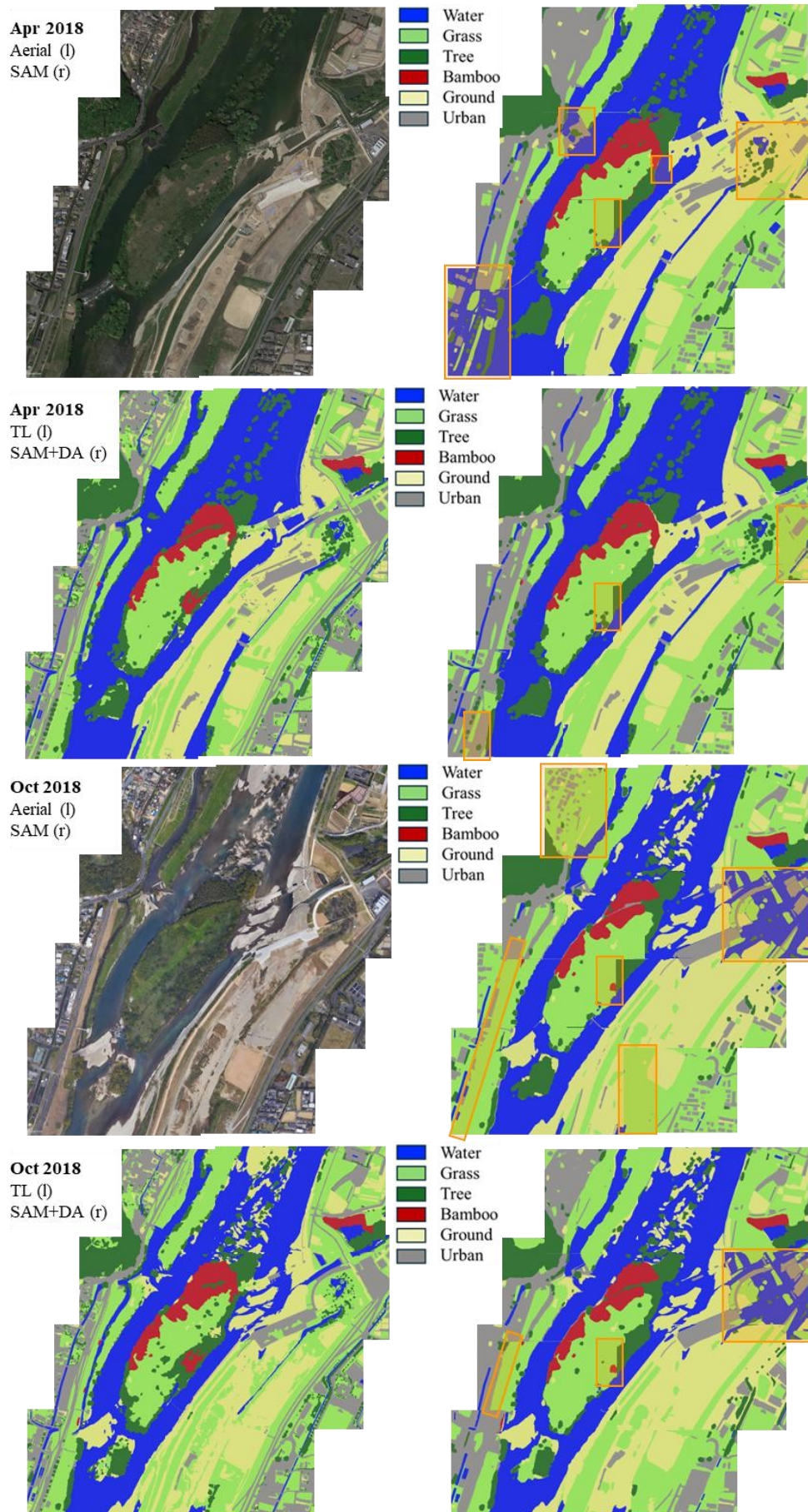


Fig.10 Result of standard (SAM) and improved LCC (SAM+DA) in April and October 2018. Limitations are indicated with orange boxes, revealed in comparison with the true label maps (TL).

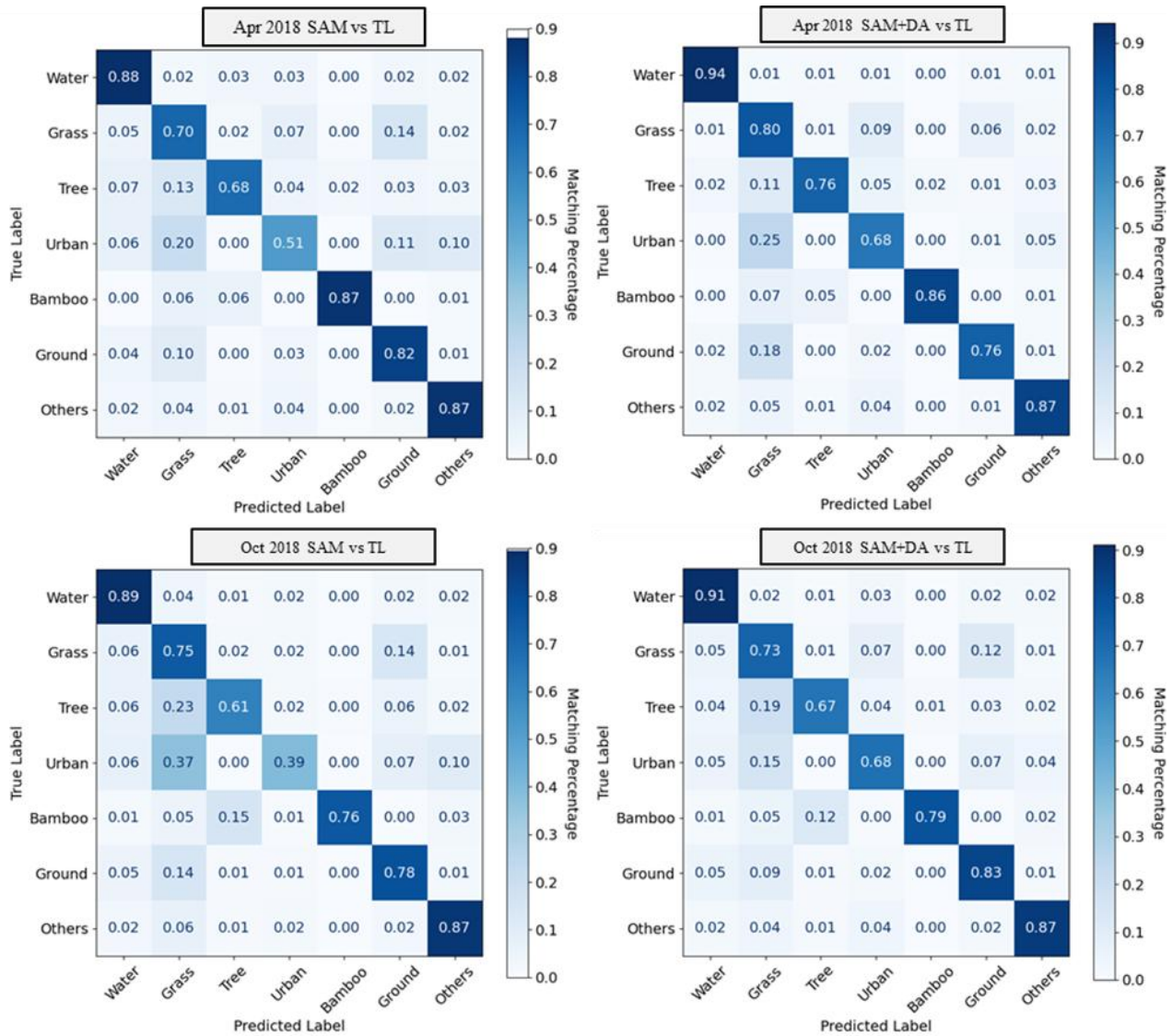


Fig.11 Quantitative comparison of the standard (SAM) and improved LCC (SAM+DA) accuracy with the true label (TL) in confusion matrices for April and October 2018.

The findings point out both the strengths and limitations of SAM as a segmentation tool. While it shows promise in handling high-contrast objects and foreground features, future work will need to address its difficulty with subtle LC distinctions and large, mixed areas. In this respect, DA is an easy-to-implement enrichment in the workflow.

The inclusion of DA overall improved the classification accuracy, particularly for urban areas, as it helped to separate large background segments more effectively. While seasonal variations slightly influenced vegetation classification performance, their impact remained relatively minor. Further refinements are necessary to minimize trade-offs between different LC categories to fully optimize SAM's potential for flood management applications.

(2) Flow measurement with STIV and Hydro-STIV

Video data for STIV was collected during the 2018 flood with CCTV cameras at Nakanohara, Nishigawara (both Asahi River) and Ninoarate (Hyakken River). Reliable video material from all cameras at the same time was not available, so we chose 24 a.m. and 6 a.m., about three hours before and after the discharge peak on July 7.

At Nakanohara, the STIV depth-averaged flow velocities show large fluctuations, especially closer to the right riverbank due to a decline in camera resolution at greater distance from the CCTV and bad lighting conditions (**Fig. 12**). The 2D analysis reference data is more plausible with high velocity where the river is deepest. Thus, the mean absolute error (MAE) for STIV is high at 0.69 m/s. However, the calculated

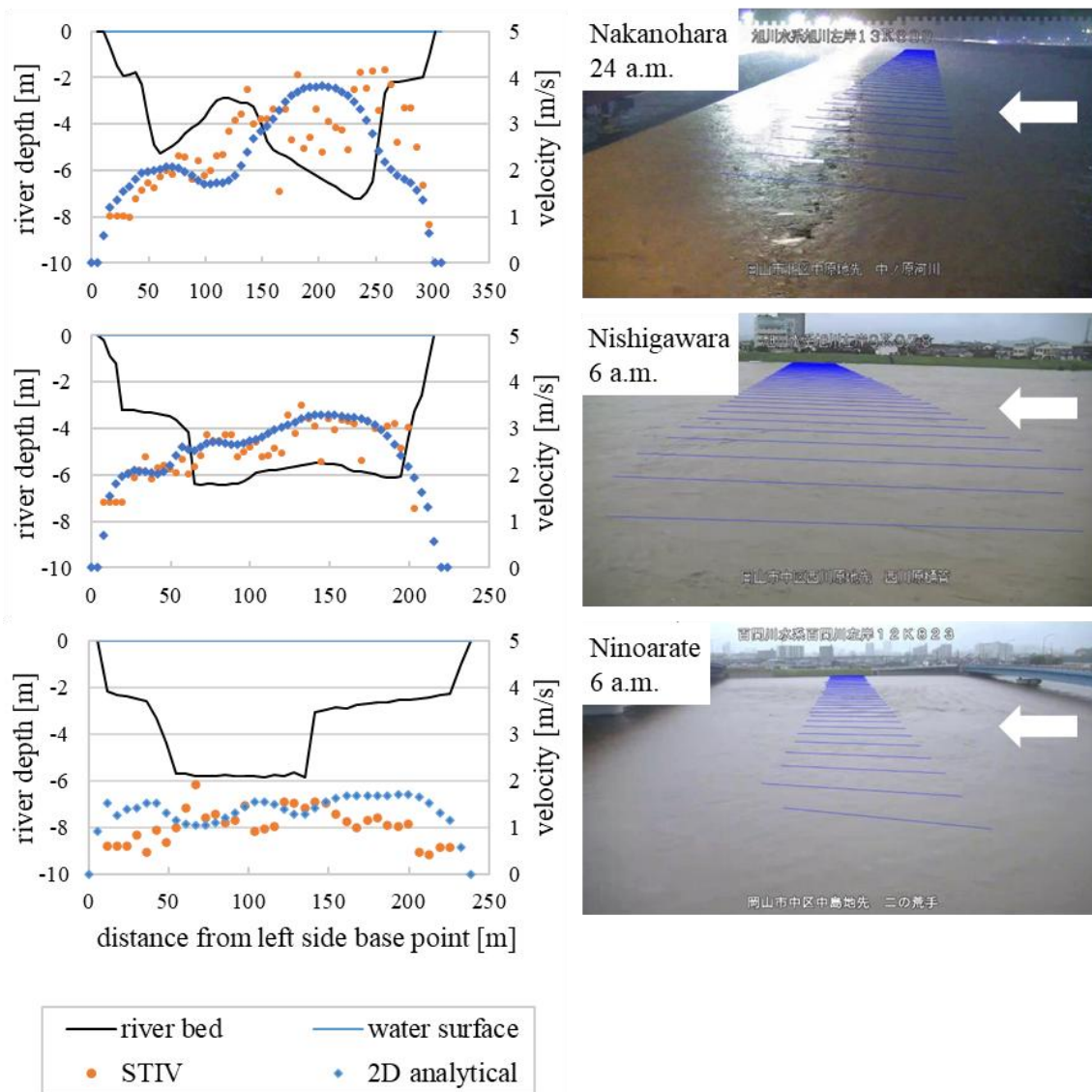


Fig.12 Depth-averaged STIV with data from CCTVs Nakanohara, Nishigawara and Ninoarate during the flood event on July 7th, 2018, before and after the peak discharge at 3 a.m. Analytical 2D model calculations were used as reference.

STIV- and analytical discharge values are similar because positive and negative deviations balance each other out.

At Nishigawara, STIV flow velocities were found to align closely with the results from the 2D hydraulic analysis, but the typical STIV fluctuations are present too. The MAE is lower at 0.27 m/s and discharge estimation differed only by 13 % compared to the 2D analysis, demonstrating the reliability of STIV in capturing floods at this location.

At Ninoarate, the STIV flow velocity was found to be less in agreement with the 2D analysis results again. The MAE is high with 0.5 m/s, reflecting the challenges of capturing slower velocities accurately. STIV underestimated the discharge by 20 %. However, the 2D analysis results at Ninoarate should be interpreted with caution due to the flow influence of the Ninoarate weir directly upstream of the CCTV. It blocks part of the river's cross-section under low

flow conditions, channeling water through a designated segment. The 2D analysis is unable to reproduce the effect of this weir, introducing flaws when using it as reference for evaluation.

The new Hydro-STIV technology was applied to a single river section in the branched Gion area of the Asahi River. The purpose was primarily to teach the participating students Hydro-STIV usage. To provide a reference and obtain riverbed information, the EM method was used. The results from the Hydro-STIV drone flights at 50 m, 60 m, and 70 m heights demonstrate the methods limitation for measuring low flow velocity accurately.

All methods produced similar parabolic surface velocity graphs, showing higher velocities in the river's center where the water depth is greatest (**Fig.13**). The MAE using the EM method as the reference were 11.57 cm/s for the 50 m H-STIV, 10.37 cm/s for the 60 m H-STIV, and 15.40 cm/s for

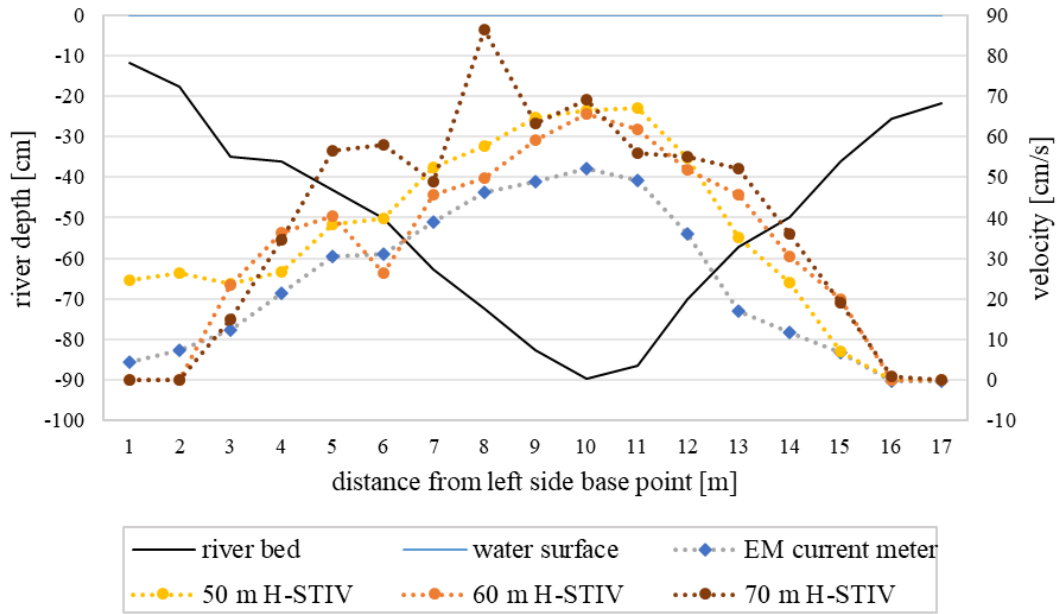


Fig.13 Depth-averaged Hydro-STIV with data from drone flights in three heights (50 m, 60 m, 70 m) during field work in Gion area under low flow condition. The electromagnetic (EM) method was used as reference.

the 70 m H-STIV. Discharge estimates showed notable overestimations compared to the EM method ($2.73 \text{ m}^3/\text{s}$): 50 m H-STIV ($3.81 \text{ m}^3/\text{s}$, 40 % overestimation), 60 m H-STIV ($3.63 \text{ m}^3/\text{s}$, 33 % overestimation), and 70 m H-STIV ($4.21 \text{ m}^3/\text{s}$, 54 % overestimation).

Among the Hydro-STIV, a flight height of 60 m yielded the most accurate results relative to the reference, especially since the 50 m flight performed poorly near the left riverbank. The 70 m H-STIV results showed the largest overestimation, due to outliers potentially caused by insufficient camera resolution at that altitude.

The EM method can be considered more reliable under low-flow conditions than Hydro-STIV. However, it has limitations, particularly human-induced flow disruptions since several students were standing in the river during measurement.

Notably, STIV and Hydro-STIV are better suited for higher flow velocities and flood conditions, where surface ripples are more distinct. But the example shows that flood will not only occur during the day under ideal conditions for STIV. The current camera setup relies on good lighting and a fixed camera viewing angle, hence improvements such as FIR cameras would help to make flood monitoring more accurate and independent from time¹⁸.

The next major flood event will enable the application of Hydro-STIV around the Diversion weir and show the advantages of Hydro-STIV over STIV.

(3) Water level forecast with RF

Random Forest flood predictions for September 2017, July 2018, and August 2021 were evaluated

through root mean square error (RMSE) and normalized RMSE (NRMSE) (**Fig.14** and **Table 2**). The 1-hour forecast yielded the most accurate results, with NRMSE values below 0.1, generally indicating good performance. Precision decreased with longer lead times (**Fig.15**). The overall prediction was most reliable in 2018-07, followed by 2021-08 and 2017-09. All water level forecasts had in common that the flood peak was predicted too early. Furthermore, RF overestimated the flood peak in 2017-09 and the first smaller peak in 2018-07 and underestimated the main peaks in 2018-07 and 2021-08. Even though the NRMSE is better in 2018-07 and 2021-08, underestimating the highest water level is more critical and should be prevented.

Table 2 NRMSE of the Asahi River flood predictions with Random Forest (1-3 hours).

	NRMSE 1h	NRMSE 2h	NRMSE 3h
2017-09	0.0696	0.1261	0.1612
2018-07	0.0515	0.0681	0.0842
2021-08	0.0612	0.0729	0.0866

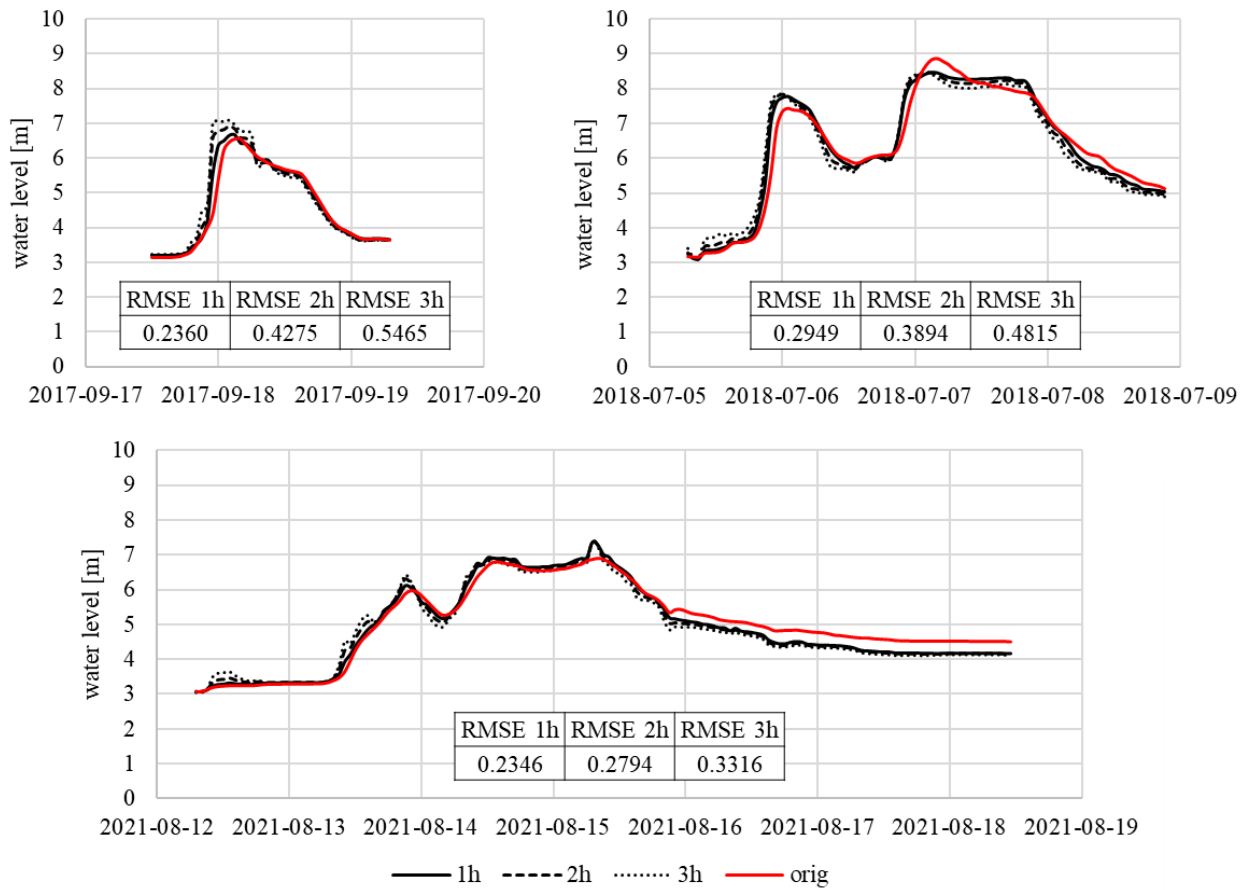


Fig.14 Flood Forecast (1-3 hours) with Random Forest and the measured water level (orig) for the flood events in September 2017, July 2018 and August 2021 (YYYY-MM-DD). Included in the diagrams are the RMSE values for each forecast lead time.

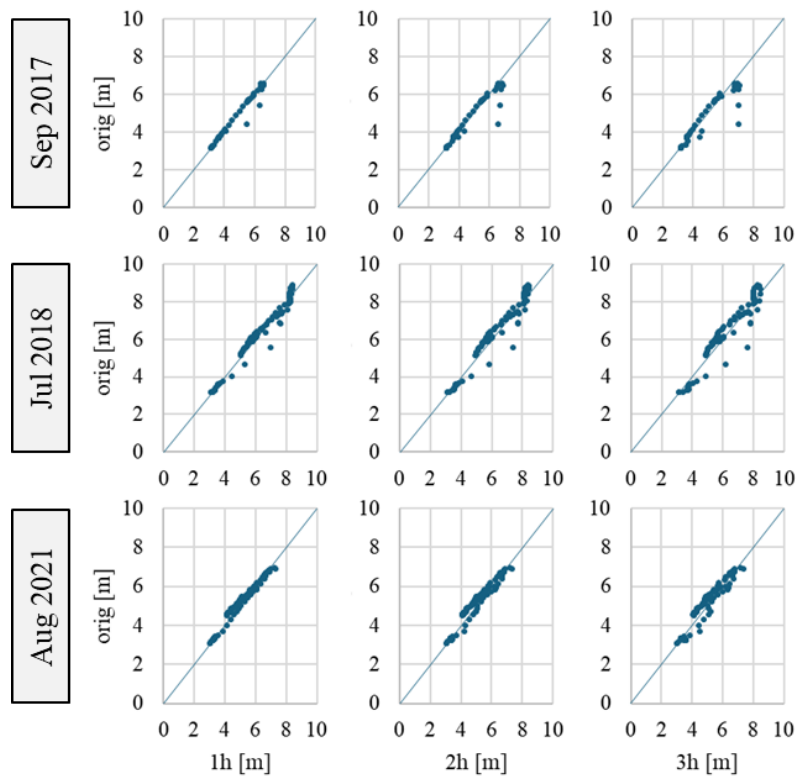


Fig.15 Scatterplots of the Flood Forecasts (1-3 hours) and the measured water level (orig) for the flood events in September 2017, July 2018 and August 2021.

(4) Integration of land cover and discharge data in RF

Integrating LC data from SAM and discharge data derived from Hydro-STIV adds both categorical and continuous variables to RF, likely enhancing its predictive capacity for floods. This combination leverages RF's ability to handle mixed data types.

Conventional discharge measurement has high uncertainties during floods, as it is usually calculated via the local statistical relationship to the water level and data of extreme floods is limited. Assuming that Hydro-STIV provides more accurate results under flood conditions^{34, 16), 17)}, forecasting with Hydro-STIV data would therefore offer a decisive advantage for accurate forecasting. Hydro-STIV discharge data could be incorporated into RF as an additional training and input variable, resulting in separate predictions of water level and discharge (**Fig.16**).

LC could be integrated as a categorical variable or boundary condition. Depending on vegetation density and flexibility, vegetation has a decisive influence on the water level. Especially under extreme water levels, for which the model is scarcely trained, vegetation parameters (e.g. roughness coefficient) could be helpful.

Once these features are added, the model would be retrained and -validated to assess how effectively the new variables improve the prediction accuracy. This expansion in feature complexity could allow the model to identify and respond to subtle interactions between land cover and hydrological patterns, a valuable development for more precise flood predictions and flood control planning.

(5) Opportunities of AI Methods for Flood Control and Sponge City Applications

The emerging field of “smart sponge cities” emphasizes the integration of advanced information technologies—such as the Internet of Things (IoT), cloud computing, artificial intelligence, and big data—to transform traditional flood mitigation into a

more proactive, intelligent, and adaptive framework⁴⁷⁾. In line with this, the integration of LCC with SAM, flow measurement with Hydro-STIV, and flood forecasting with RF offers a promising combination for analyzing river segments.

Despite this potential, current limitations exist. SAM's improved segmentation performance, while equally good compared to a green lidar system (GLS)⁴¹⁾, still falls short compared to costly LCC methods like airborne LiDAR topo-bathymetry (ALB)⁹⁾. Moreover, manual LC labeling remains time-intensive, which presents an obstacle to large-scale implementation. Nonetheless, as aerial image resolutions improve and SAM undergoes further model training, LCC with SAM is expected to offer increasingly accurate, automated classifications. As an open-source tool, it comes with financial benefits too.

LC influences surface runoff due to its infiltration and water storage capacity, as well as water levels during floods due to its surface roughness. Particularly vegetation changes rapidly, which makes regular updates essential for maintaining reliable hazard maps. Additionally, LC maps can provide valuable insights into some sponge city indexes like water surface ratio and green coverage. Estimations regarding the water absorption rate would further enable the specification of the sponge green rate (**Fig 1**). Those indicators support decision-making for optimized implementation of sponge city elements like green infrastructure.

Traditional contact-based methods for measuring river discharge face limitations in both accuracy and continuity during extreme flood conditions. In contrast, Hydro-STIV presents a cost-effective and relatively accurate solution for flood monitoring. Basic hydrological data is fundamental for river disaster countermeasure planning¹⁷⁾ and thus sponge cities. Still, Hydro-STIV relies on good lighting conditions or compensating FIR cameras. In Okayama City, Hydro-STIV could play a crucial role

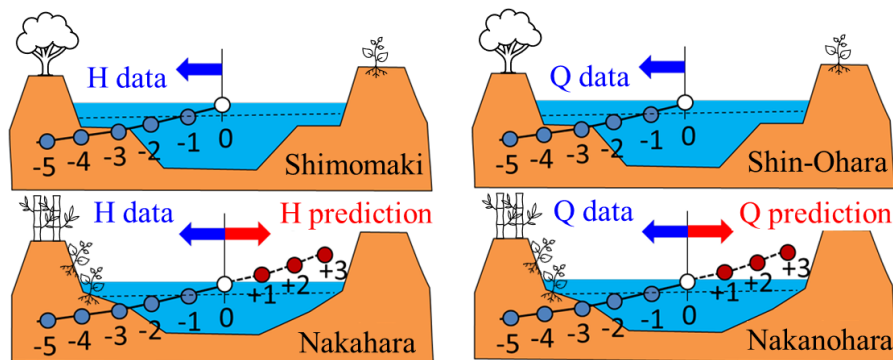


Fig.16 Possible water level (H) and discharge (Q) prediction (1-3 hours) for Nakahara Station and Nakanohara using previous data (0–5 hours) from gauging stations and CCTV cameras respectively (Fig.3). Vegetation information could be used as boundary conditions as it affects water levels, especially under extreme conditions.

in controlling the discharge diversion. Together with LC data, this information can help to identify the need for maintenance measures, such as vegetation clearing.

Flood prediction using RF has shown to be promising for short-term water level forecasts. Accurate flood prediction is vital for proactive flood management. Within the context of sponge cities, this could enable controlled flooding of designated areas, such as rice paddies, to mitigate flood risks. However, a 3-hour lead time limits active flood control measures, such as issuing alerts or making operational decisions. Current practices for long-term forecasting typically rely on hydrological or numerical weather prediction models, which consider seasonal patterns and historical data. RF can be adapted for similar long-term forecasting by incorporating these factors and increasing the training dataset. Alternatively, LSTM models, designed to better capture time-series dependencies, could be employed to address the temporal aspects of flood events more effectively, providing an additional layer of robustness to flood forecasting⁴⁹.

In summary, the AI-based tools examined in this study contribute to key sponge city framework indexes and support decision making in sponge cities.

4. CONCLUSION

While other studies used SAM for LCC, Hydro-STIV for river flow estimation, and RF for hydrological prediction, this study underscores the combined potential of those AI-driven methods to enhance flood control in urban areas, particularly in the evolving context of smart sponge cities. By leveraging these tools, flood risk management can shift from reactive to proactive, combining real-time monitoring with predictive modeling to mitigate the impact of extreme weather events more effectively.

The integration of SAM-based LCC and Hydro-STIV-derived discharge data into the RF model could prove highly valuable. As the model offers both categorical and continuous variables, it can capture the complex interactions between land cover, flow dynamics, and flood risk. This integration will likely improve predictive accuracy and emphasizes RF's flexibility in handling diverse data types, enabling more nuanced analysis of flood risk within river basins.

Despite these advances, challenges remain, including SAM's limited precision to segment certain land cover types in its general segmentation configuration and the time-intensive nature of manual LCC labelling. However, the ongoing development of higher-resolution aerial imagery and

training of SAM will progressively address these limitations. Additionally, the integration of DA in the process led us one step closer to the desired LCC accuracy.

Hydro-STIV, as an advanced non-contact, image-based discharge measurement, provides surface flow reading estimates under favorable factors and holds promise for use during floods. However, its dependency on image quality and brightness conditions, as well as its limitations in measuring low or multi-directional flows indicate that adapted variants—such as dual-camera setups or infrared cameras—may be required for consistent, long-term application. A real-time version of Hydro-STIV called 'Hydro-STIV Real Time' has already been developed.

Finally, the application of RF for flood prediction shows promise for short-term, real-time forecasts, though the 3-hour lead time used in this study remains insufficient for emergency response measures. Future adaptations incorporating seasonal patterns and larger training datasets could extend RF's applicability for longer-term forecasts.

Collectively, these findings underscore AI's transformative role in flood control management, with advancements in LCC, discharge measurement, and predictive modeling. The three AI tools examined in this study therefore contribute to key sponge city framework indexes and support decision making in sponge cities.

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